Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

# Sound Localization Robot, Team 5

## Final Presentation

**Samuel Lackey, Sanjana Pasumarthi, Lucas Kettler, Joshua Teele**

**TA: Muhammad Bin Mughees**
**Sponsor: Dr. Moreira**

# Project Scope

*Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)*

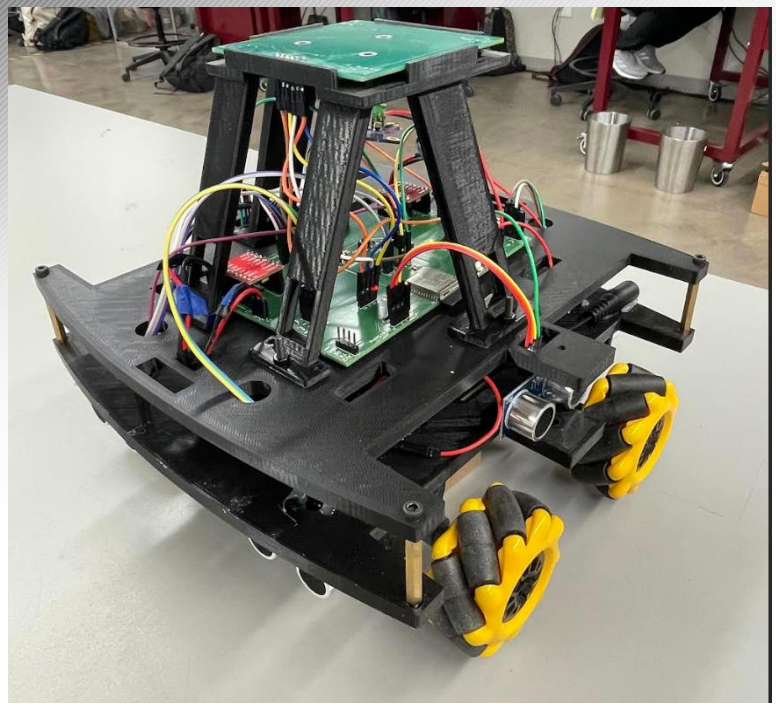
- Keeps **first responders safe** by entering hazardous or toxic areas first.
- **Reduces search and rescue time** by avoiding exploring unnecessary paths
- **Reliable in low-visibility** or high-stress conditions (smoke, debris, noise).

**Problem:** First Responders lack reliable methods for navigating dangerous environments, where visibility is limited, putting both their safety and the victims at risk.

**Our Solution:** Building a Sound Localization robot that can hear distress calls, pinpoint their direction, and move toward them autonomously, allowing first responders to focus on saving lives instead of risking theirs.

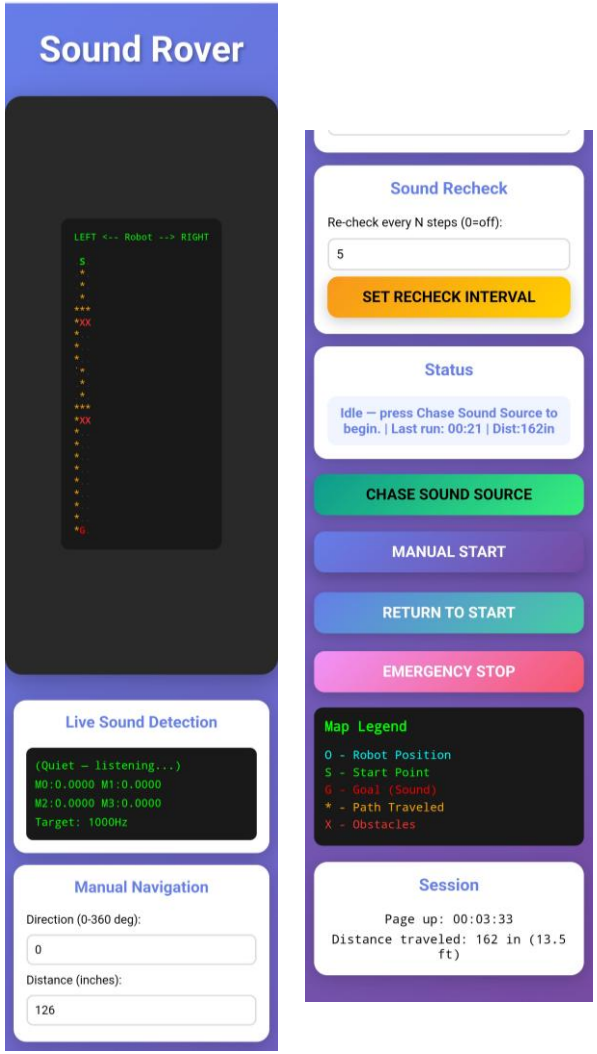
*A smarter, safer, and faster way to locate victims, all while protecting rescuers and saving more lives.

# Integrated System Diagram:



Website generated through internal Wi-Fi, data through TCP

User Commands: Emergency Stop, Origin




User Interface – viewed from any device with Wi-Fi capabilities

Fully Integrated System Design:

- Integrated PCB (w/ ESP32) for motor controls, power, distance sensors, and navigation
- MEMS Microphone PCB on top layer

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# Navigation Software Subsystem
Lucas Kettler

```
// grid[y][x]: 0 = free space, 1 = obstacle
std::vector<std::vector<int>> grid = {
    {0, 0, 0, 0, 1},
    {1, 1, 0, 1, 0},
    {0, 0, 0, 0, 0},
    {0, 1, 1, 1, 0},
    {0, 0, 1, 0, 1}
};
```

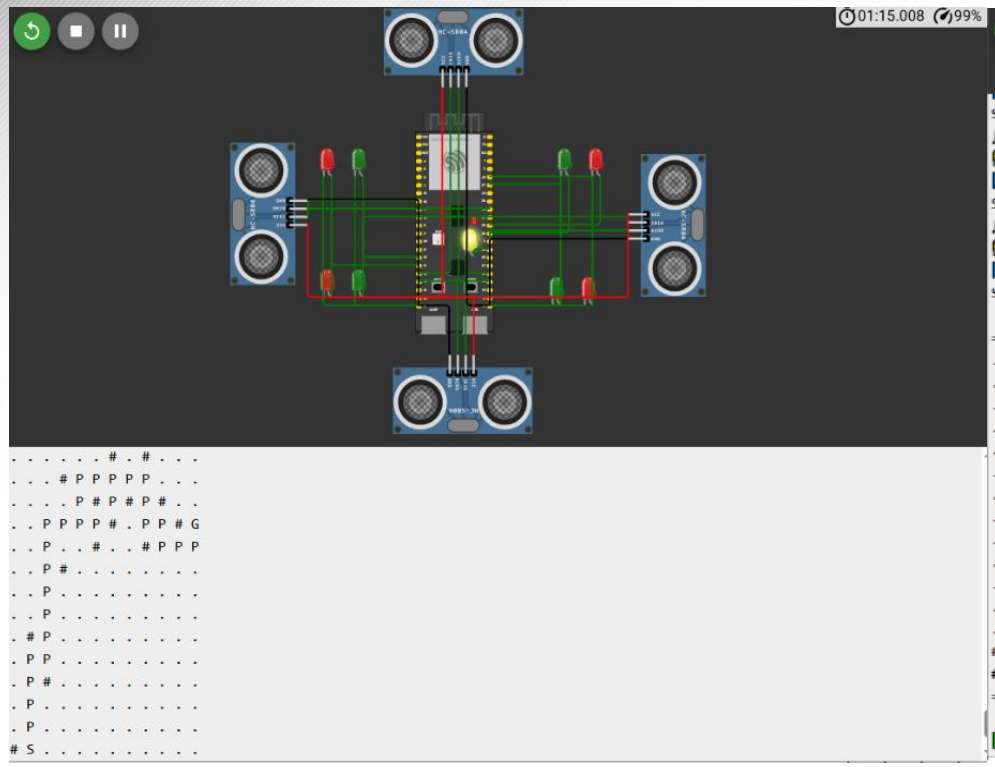
```
Found path! Length = 15
(8,4)
(8,3)
(8,2)
(8,1)
(8,0)
(7,0)
(6,0)
(5,0)
(4,0)
(3,0)
(2,0)
(2,1)
(2,2)
(1,2)
(1,3)
(1,4)

=== MAP WITH PATH ===
..........
..PPG.....
PPP####...
P.#....#..
P#......#.
P#......#.
P#......#.
P#......#.
PPPPS.....
..........
====================

PS D:\C++>
```

```
1   ##########
2   #S.......#
3   #.######.#
4   #.#....#.#
5   #.#....#.#
6   #.#..G.#.#
7   #.######.#
8   #........#
9   #........#
10  ##########
11
```

```
PS D:\C++>
PS D:\C++>  & 'c:\Users\User\.vscode\extensions
Out-xuohrxau.vyp' '--stderr=Microsoft-MIEngine-
NO PATH FOUND
PS D:\C++>
```

Only functioned with known environments

# Navigation Software Subsystem
Lucas Kettler



No real world implementation at that time

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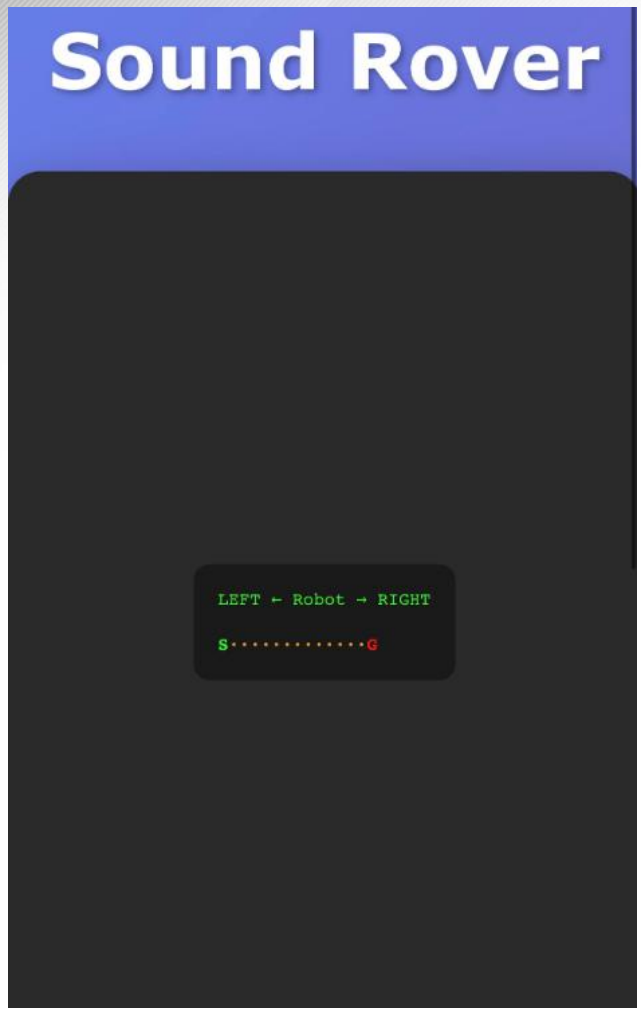
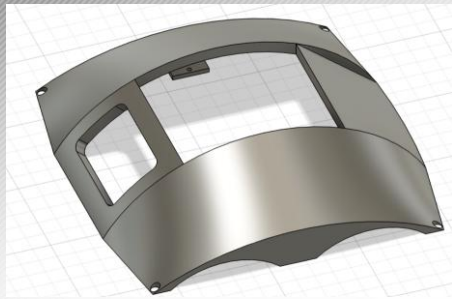
# Navigation Software Subsystem

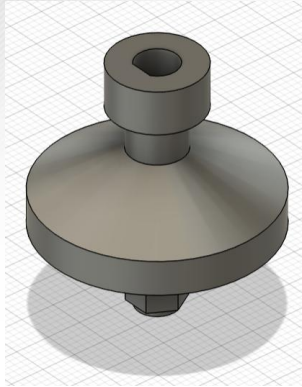
Lucas Kettler



# Motor Controls & Chassis Design & UI Subsystem

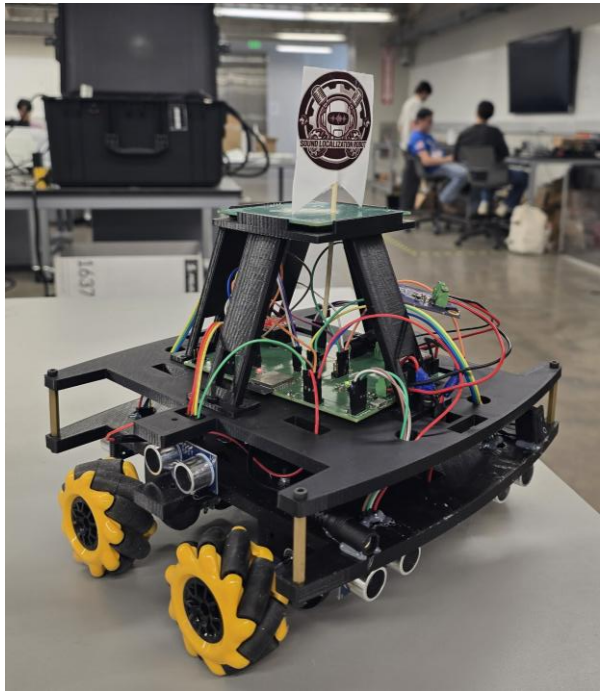
Sanjana Pasumarthi

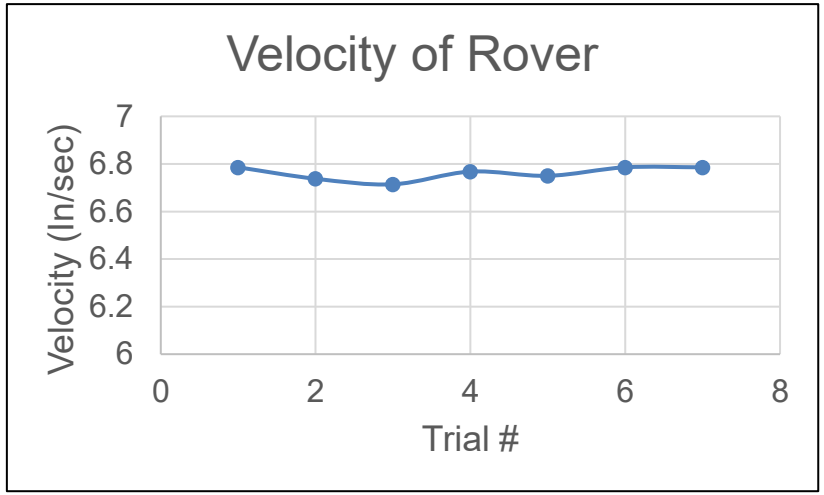




- Physical Design of Rover
  - 2 layer Chassis Design
  - Strong Wheel axle
  - Dome Casing

- Rover System
  - TCP socket generated from ESP Wi-fi
  - Webpage consists of multiple commands for user

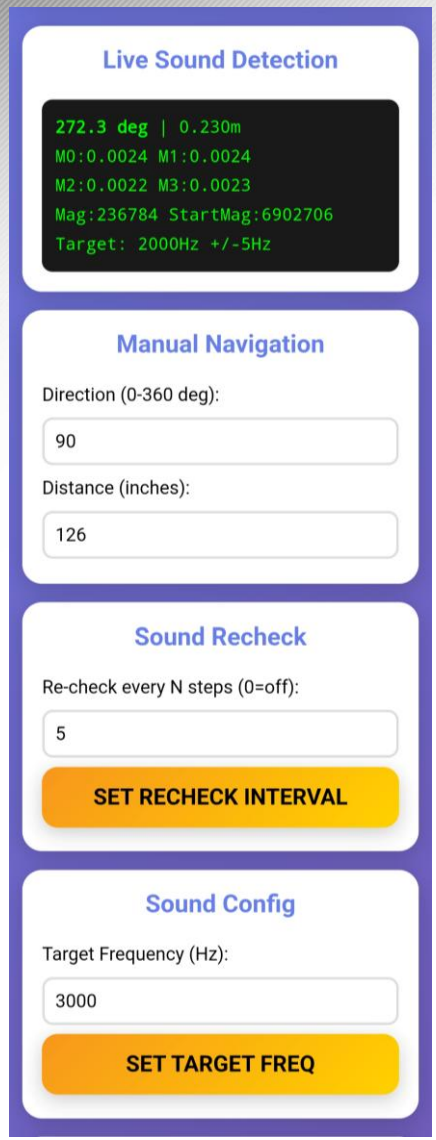


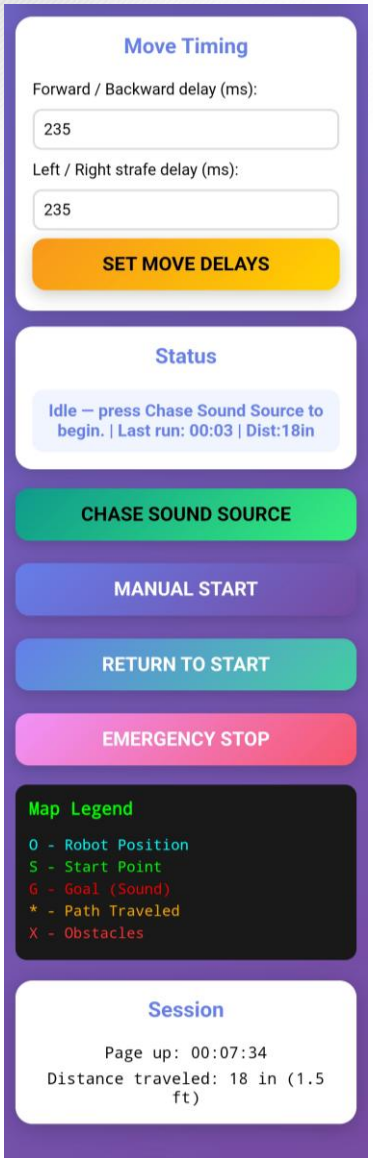



Avg: 6.76 In/sec

# Motor Controls & Chassis Design & UI Subsystem

Sanjana Pasumarthi




| Test | Method used | Expected Result | Pass? |
|---|---|---|---|
| Strong Axle for quick response | Different design of axle, that takes into consideration the wheel & torsion | Sturdy & thicker shaft | Yes – observed breaking capacity |
| WebApp must display commands for user | Use ESPs internal Wi-Fi to set up a TCP socket | User-friendly website to control rover from any distance | Yes – multiple settings of rover can be controlled from the web |
| Accurate Robot Movement | Multiple Series of Tests for robot movement | Robot follows user commands properly | Yes – Robot responds to commands given by user, but movement is off by 1 inch for every 20ft moved |

- Challenges vs Solutions

- Defects with motor movement, so calibrated motors respective to each direction to get proper movement

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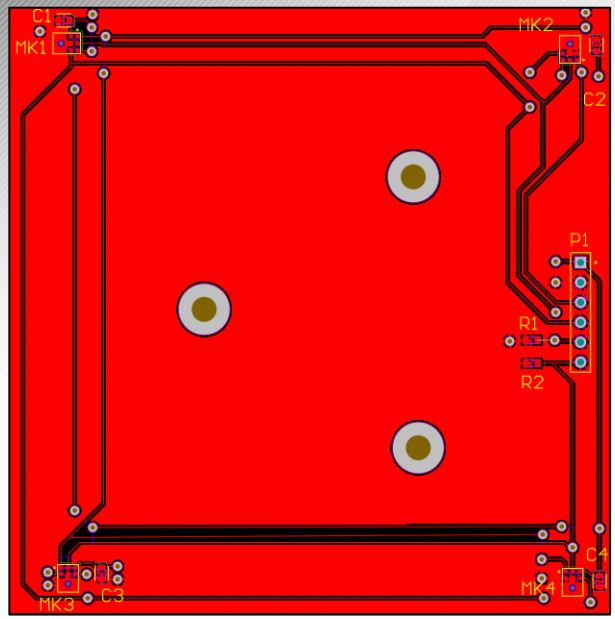
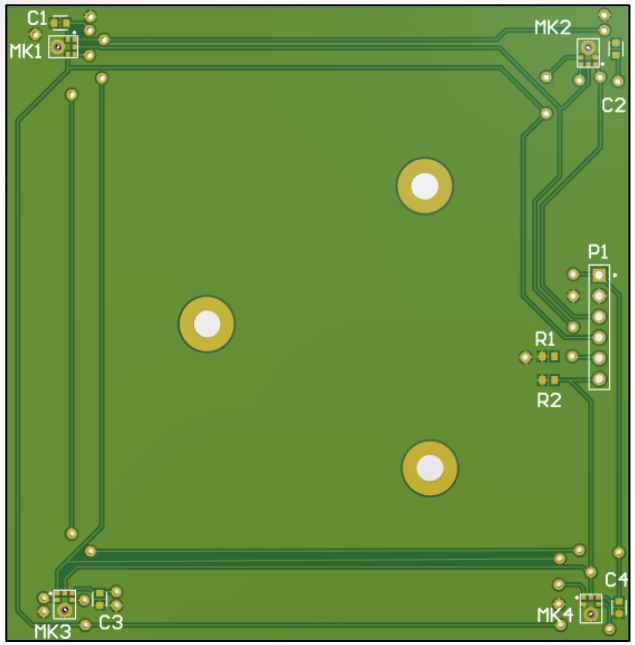
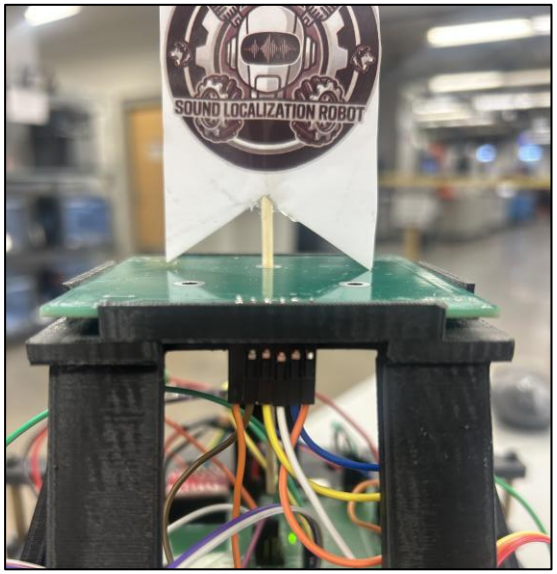
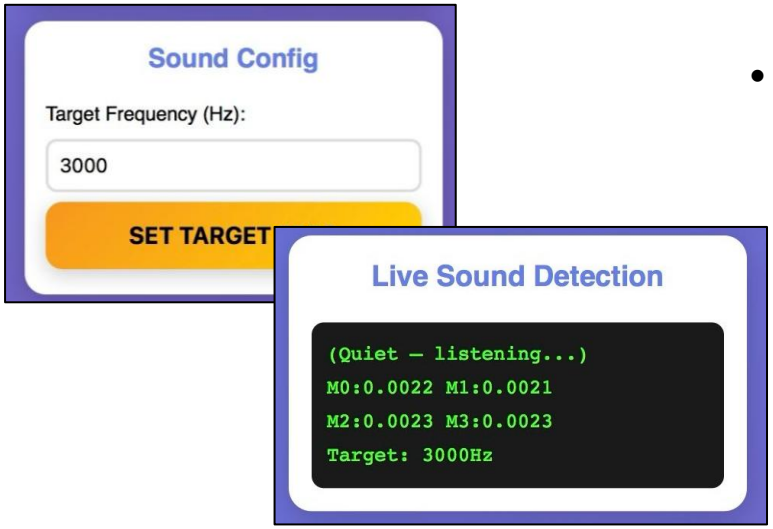
# Input Signal Subsystem
Samuel Lackey



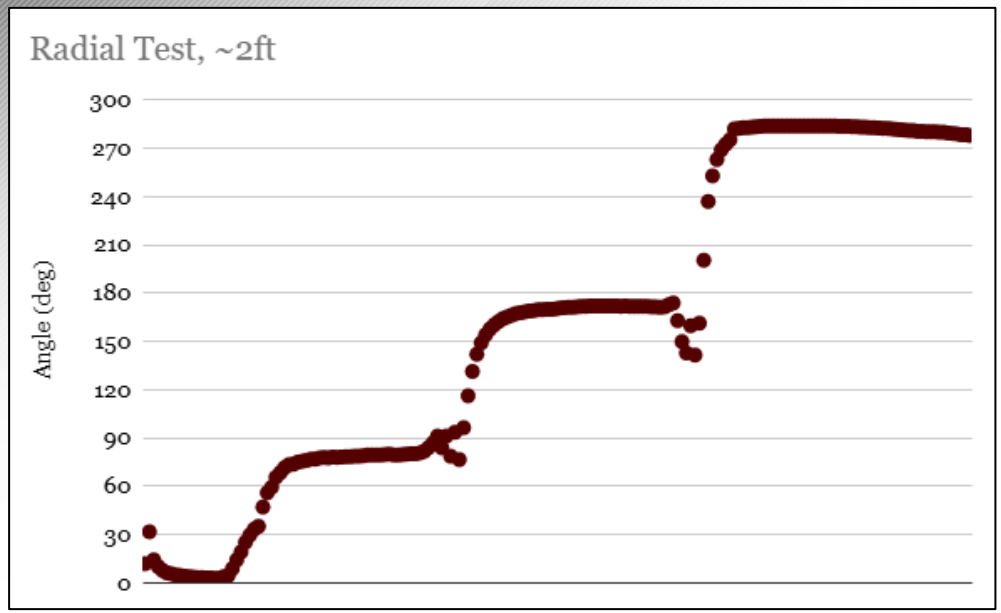

## Challenges:

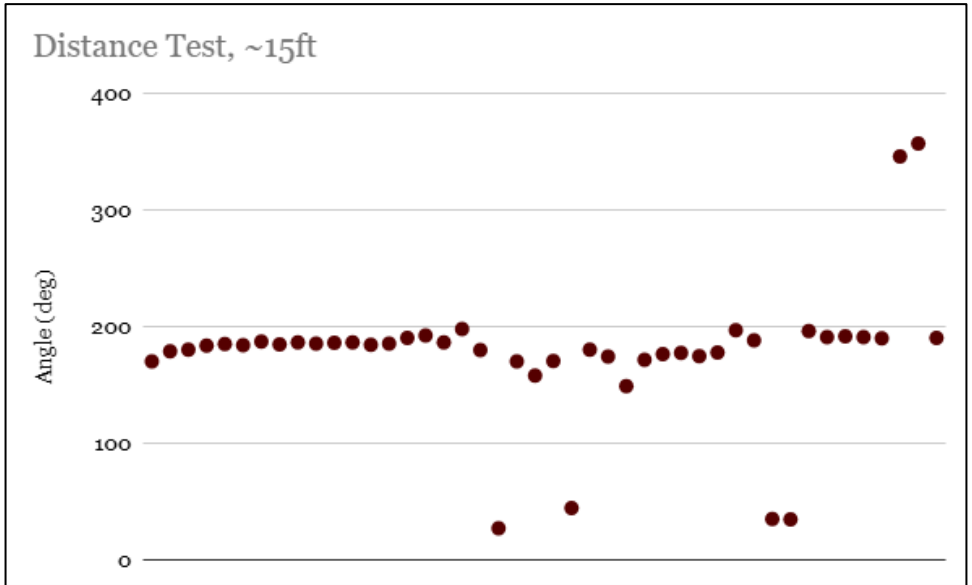
- Difficult to do sound localization in a noisy room
- Distance is physically difficult

## Solution:

- GCC-PHAT Algorithm
- Estimate Distance moved and relative location of goal

# Input Signal Subsystem
Samuel Lackey



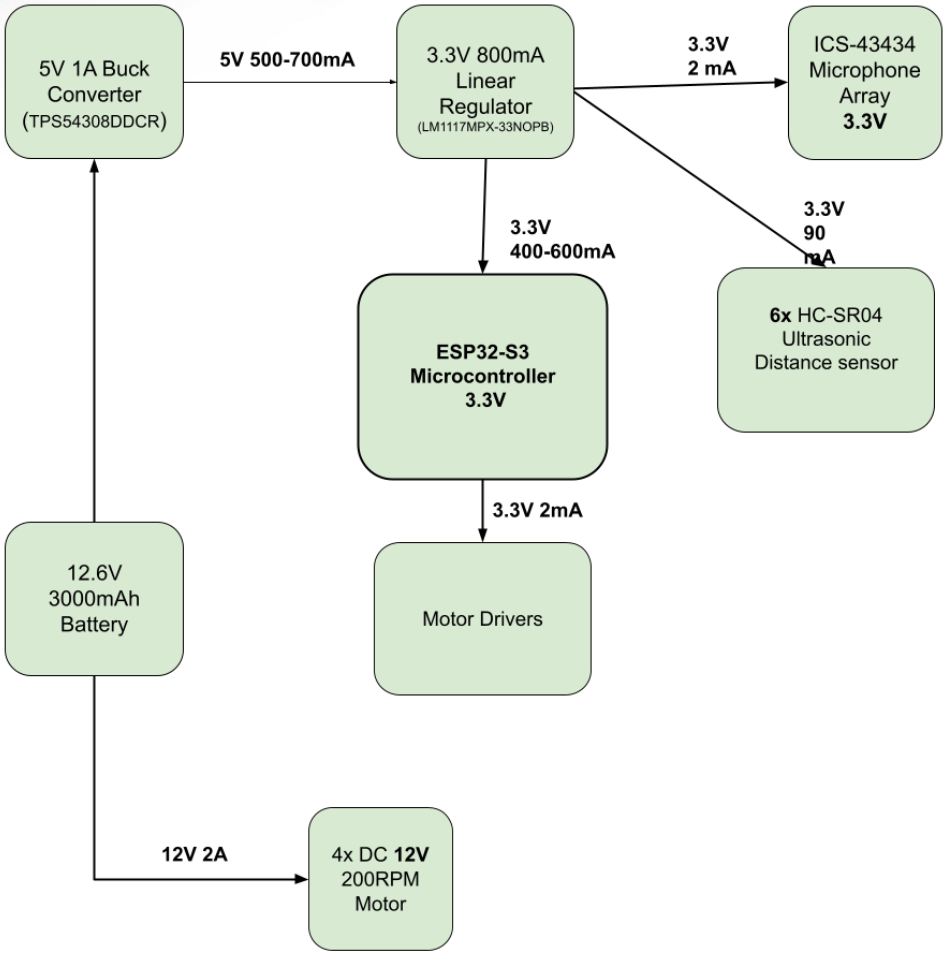




| Test | Pass? |
|---|---|
| Microphone gives radial angle < 15° | 0° - ✓<br>90° - ✓<br>180° - ✓<br>270° - ✓ |
| Microphone gives radial angle < 15° - With Background Noise | 0° - ✓<br>90° - ✓<br>180° - ✓<br>270° - ✓ |
| Angle is found at a distance of 15ft | ✓ |

# Power & Distance Sensor & MCU Subsystem

Joshua Teele

## Diagram of Power Flow in Whole System



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# Power & Distance Sensor & MCU Subsystem

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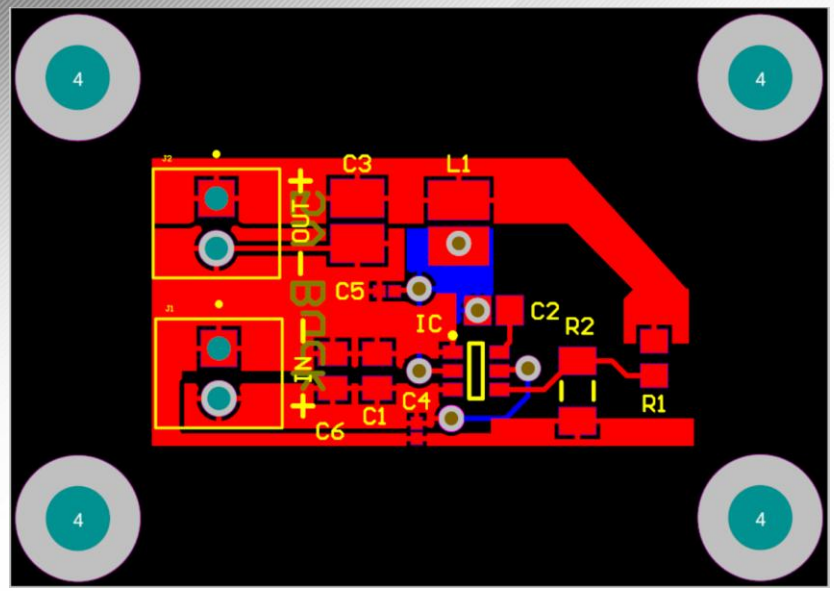
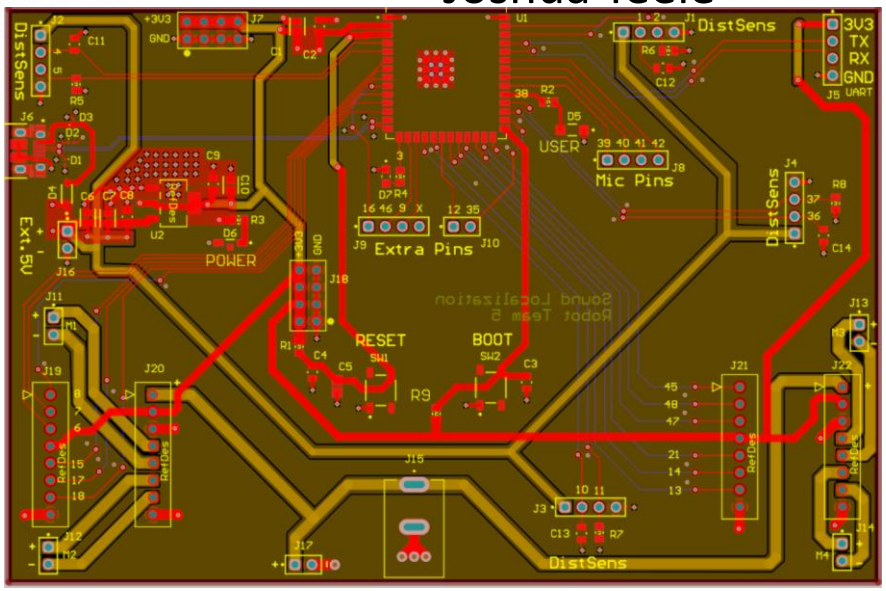
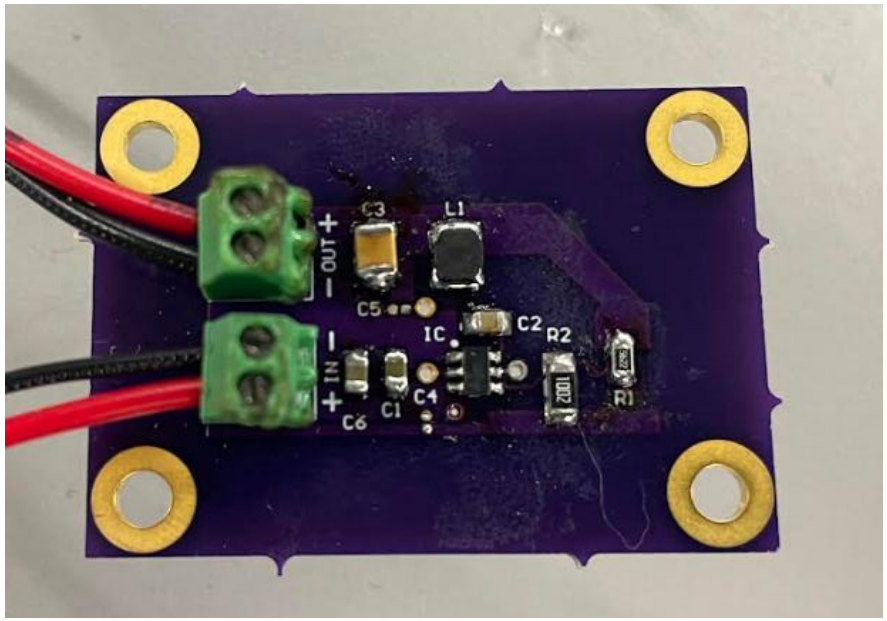
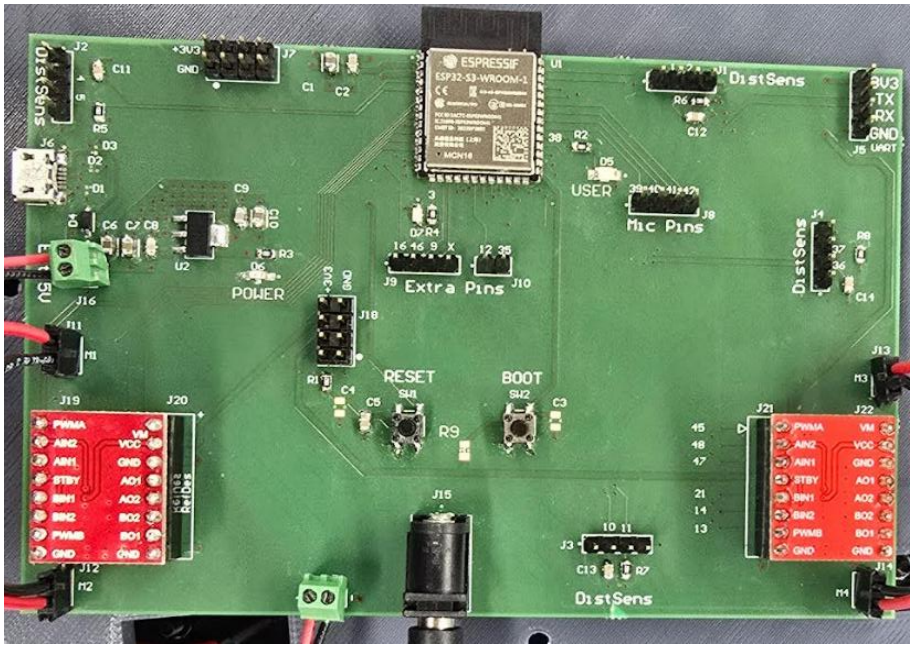


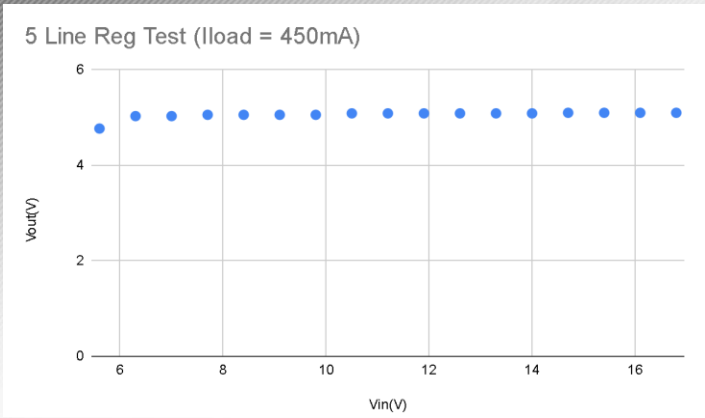



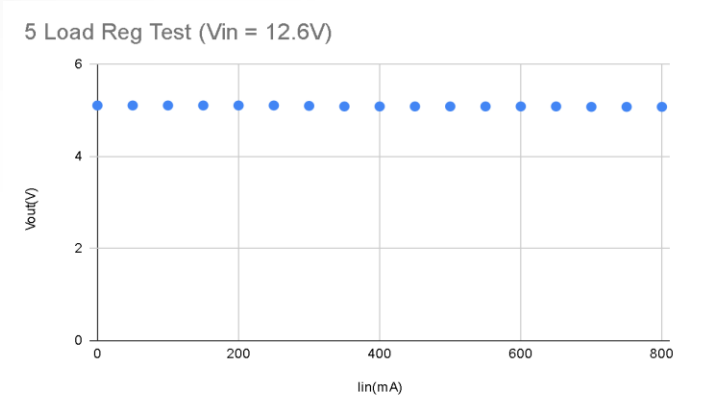



# Power & Distance Sensor & MCU Subsystem

Joshua Teele




**Line Regulation**

$$= \frac{\Delta VL}{\Delta Vi} * 100\%$$

$$= 3.14\%$$

**Load Regulation**

$$= \frac{V_{NL} - V_{FL}}{V_{FL}} * 100\%$$

$$= 0.591\%$$

| Test | Distance used | Expected Result | Pass? |
|---|---|---|---|
| Sensor outputs high when distance < 25.4 cm | 15 cm | High on all sensors | Sensor 1 - ✓<br>Sensor 2 - ✓<br>Sensor 3 - ✓<br>Sensor 4 - ✓ |
| Sensor outputs low when distance > 25.4 cm | 35 cm | Low on all sensors | Sensor 1 - ✓<br>Sensor 2 - ✓<br>Sensor 3 - ✓<br>Sensor 4 - ✓ |
| What happens at the boundary for sensor | 25.4 cm | High on all sensors | Sensor 1 - High<br>Sensor 2 - High<br>Sensor 3 - High<br>Sensor 4 - High |
| Slightly under boundary | 24.5 cm | Low on all sensors | Sensor 1 - ✓<br>Sensor 2 - ✓<br>Sensor 3 - ✓<br>Sensor 4 - ✓ |
| Slightly over boundary | 26.4 cm | High on all sensors | Sensor 1 - ✓<br>Sensor 2 - ✓<br>Sensor 3 - ✓<br>Sensor 4 - ✓ |
| Multiple sensors | 25.4 cm | High on all combinations | ✓ |

Challenges:

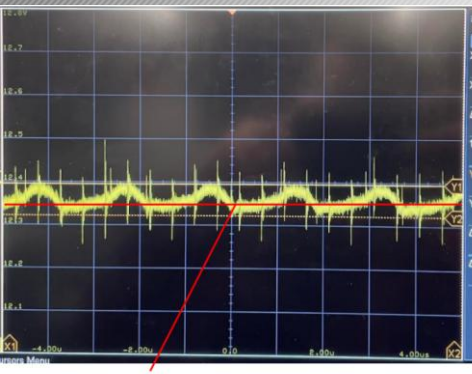
- Placement of distance sensors to avoid obstacles

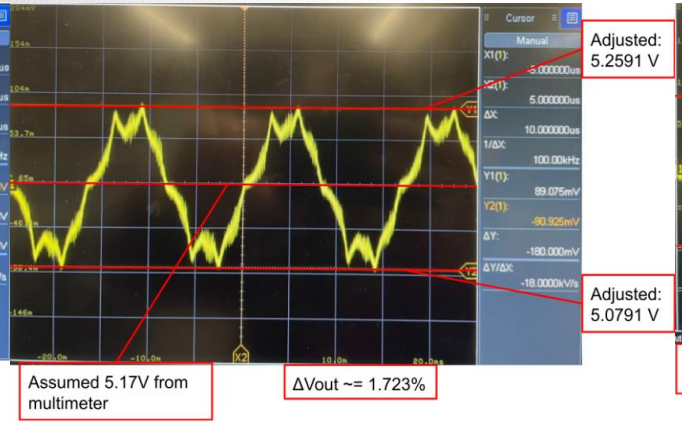
Solutions:

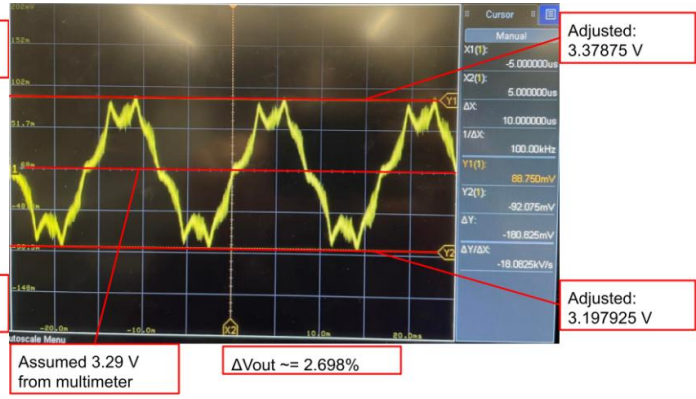
- Put on front and back where it can catch blind spots

# Power & Distance Sensor & MCU Subsystem

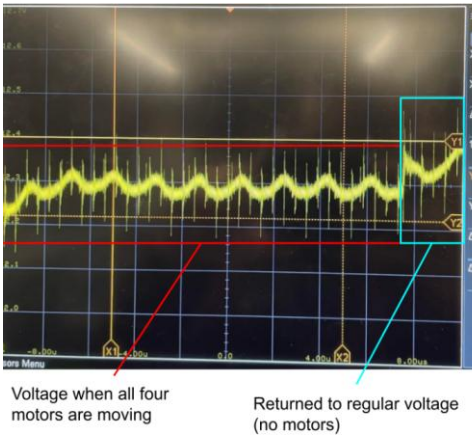
Joshua Teele

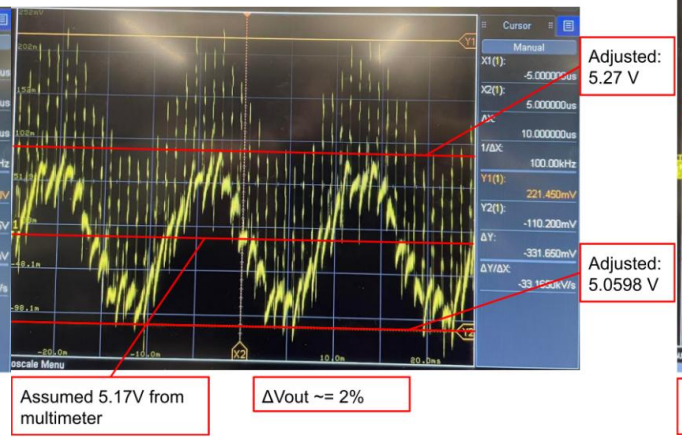


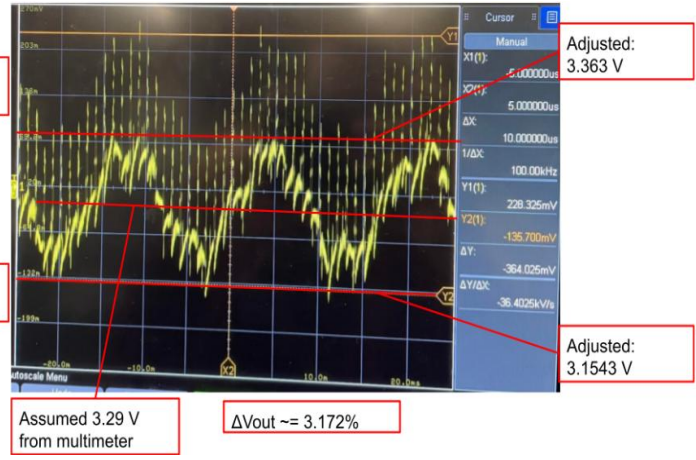




No motors on, only running ESP

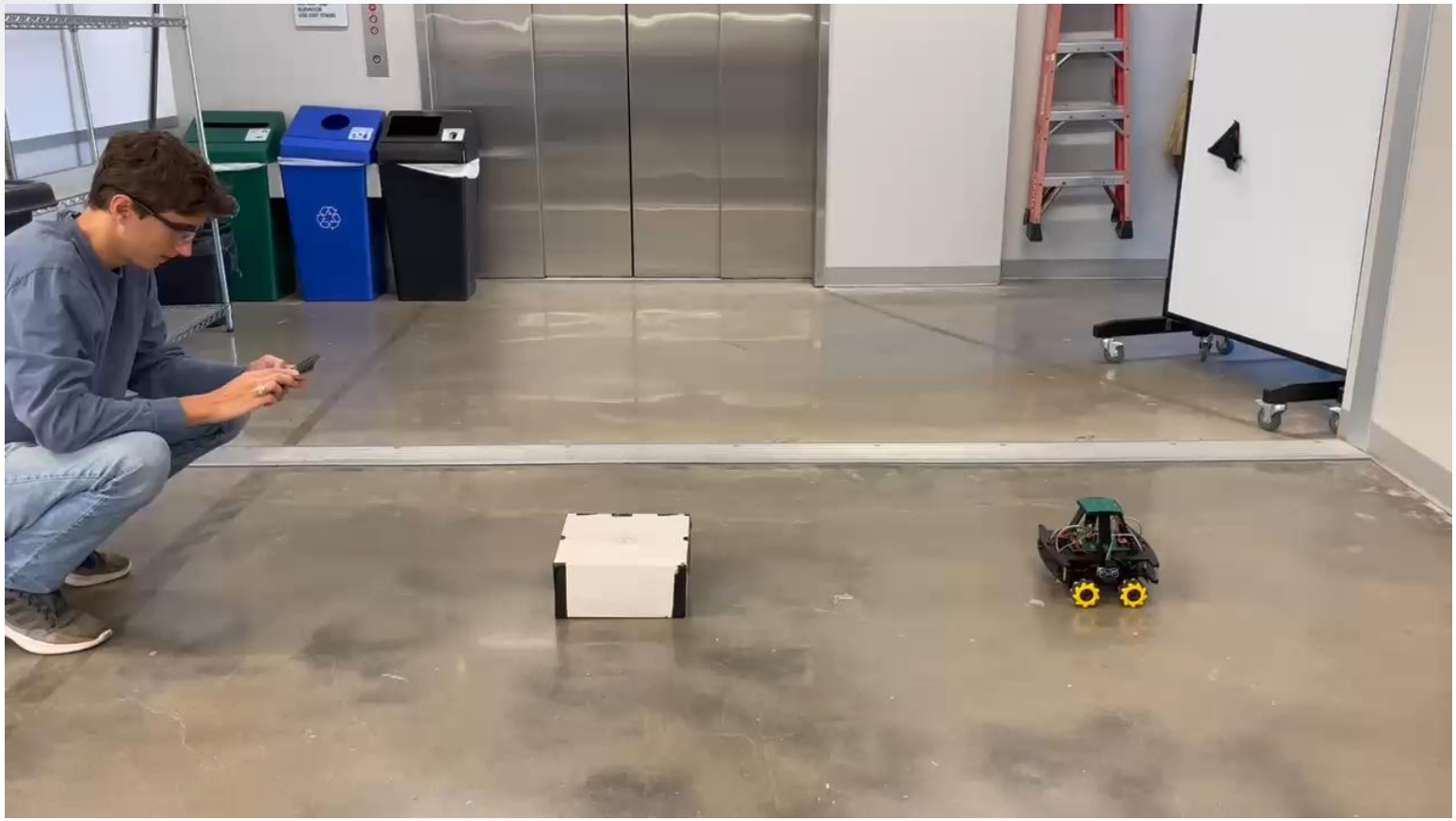


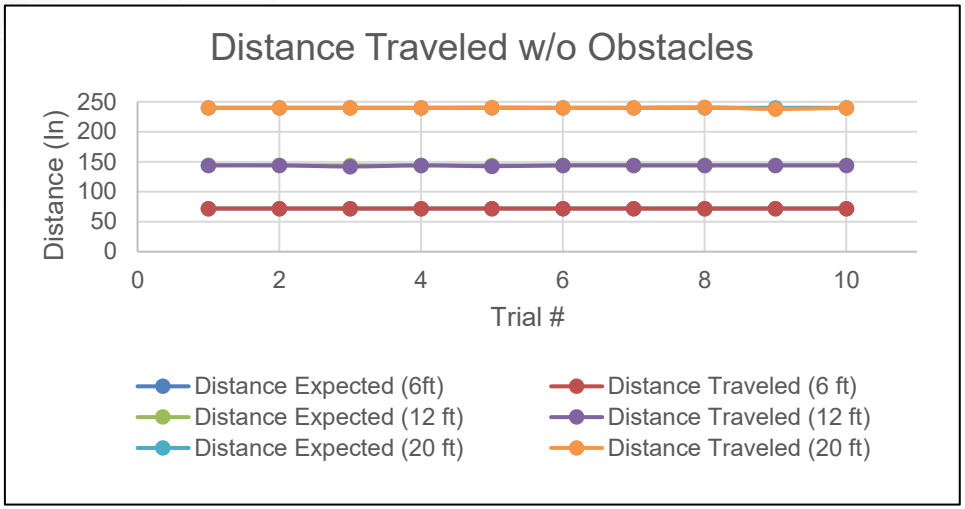

All motors on and running ESP

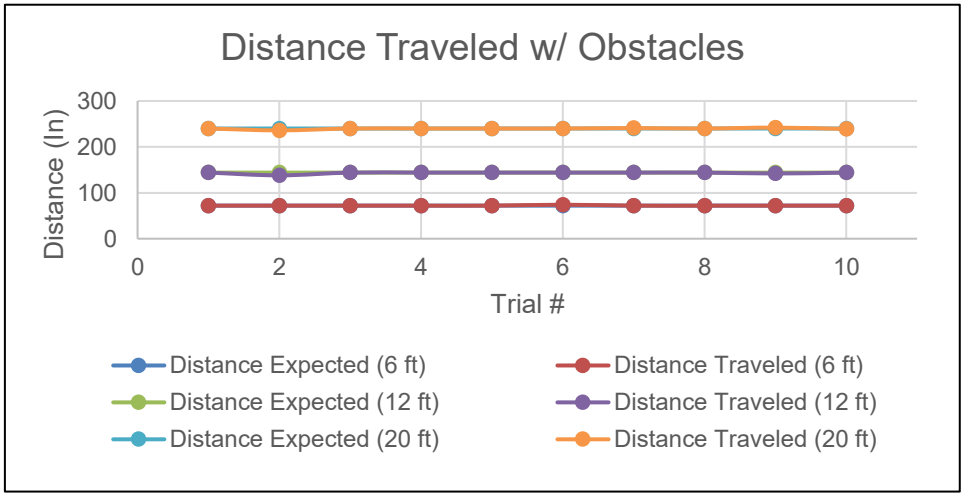
# System



# System Testing

| Test | Results |
|---|---|
| No Obstacles<br>ME: +/- 3.83 % | 6ft – 10/10 |
| | 12ft – 8/10 |
| | 20ft – 7/10 |
| With Obstacles<br>ME: +/- 5.98 % | 6ft – 9/10 |
| | 12ft – 8/10 |
| | 20ft – 7/10 |




All tests completed using a 2kHz sound source, and all 4 sides of robot tested

# Conclusions

**Changes from 403:**

- Additional Distance Sensors
- No 3.3 V Buck Converter
- No Hall effect sensors
- Using designed PCB Mic Array

**Current Status:**

- Integration Complete
- Validation Complete
- Finishing Touches such as addressing slight motor drift, and map display

# Thank You,
# Any Questions?

Sound Localization Robot 5 – Team 23

Lucas Kettler, Samuel Lackey, Joshua Teele, Sanjana Pasumarthi